

Software Engineering 11–12 (2022)

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Software Engineering 11–12 (2022)

Overview

Syllabus overview

Organisation of Software Engineering 11–12

The Year 11 course provides students with opportunities to develop and apply an understanding of the fundamental elements involved in creating software.

The Year 12 course provides students with opportunities to extend their knowledge, understanding and skills in the development of software. A major software engineering project provides students with the opportunity to further develop project management skills.

Figure 1 shows the organisation of Software Engineering 11–12.

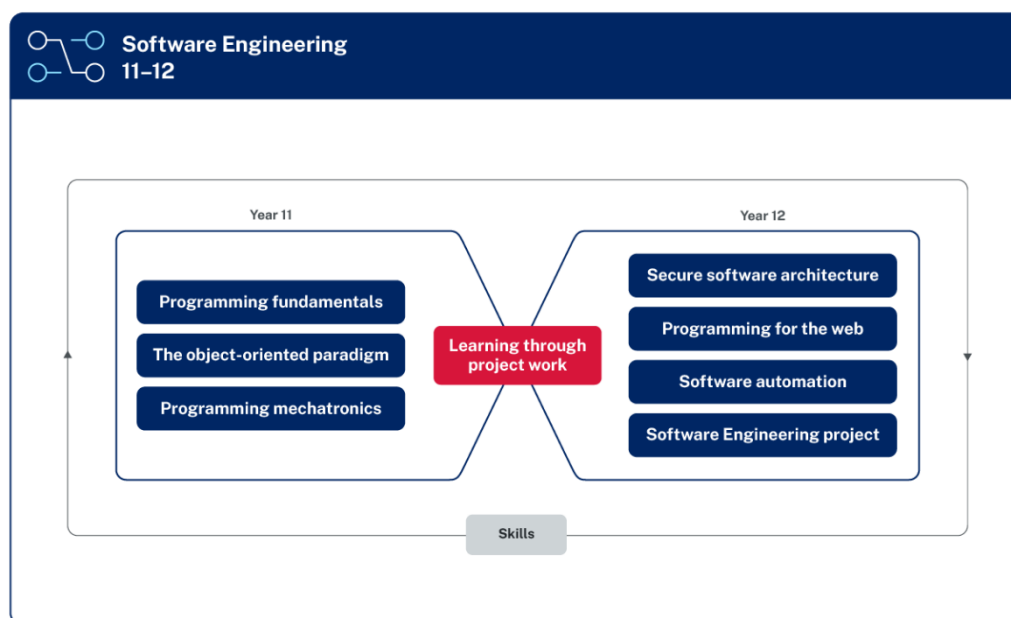


Figure 1: The organisation of content for Software Engineering 11–12 Syllabus

Image long description: This is a diagram outlining the organisation content for the Software Engineering 11–12 Syllabus. Content is listed in boxes, Programming fundamentals, The object-oriented paradigm, and Programming mechatronics fall under Year 11 on the left. Secure software architecture, Programming for the web, Software automation and Software Engineering project fall under Year 12 on the right. In the middle is Learning through project work, with lines coming from this box and encircling the content. Surrounding all content of the diagram is a line which is joined at the bottom by a box labelled, Skills. This demonstrates that Skills is being both developed and used in all focus areas.

Protocols for collaborating with Aboriginal and Torres Strait Islander Communities

NESA is committed to working in partnership with Aboriginal Communities and supporting teachers, schools and schooling sectors to improve educational outcomes for young people.

It is important to respect appropriate ways of interacting with Aboriginal Communities and Cultural material when teachers plan, program and implement learning experiences that focus on Aboriginal and Torres Strait Islander Priorities.

Indigenous Cultural and Intellectual Property (ICIP) protocols need to be followed. Aboriginal and Torres Strait Islander Peoples' ICIP protocols include Cultural Knowledges, Cultural Expression and Cultural Property and documentation of Aboriginal and Torres Strait Islander Peoples' identities and lived experiences. It is important to recognise the diversity and complexity of different Cultural groups in NSW, as protocols may differ between local Aboriginal Communities.

Teachers should work in partnership with Elders, parents, Community members, Cultural Knowledge Holders, or a local, regional or state Aboriginal Education Consultative Group. It is important to respect Elders and the roles of men and women. Local Aboriginal Peoples should be invited to share their Cultural Knowledges with students and staff when engaging with Aboriginal histories and Cultural Practices.

Course description

Software Engineering 11–12 provides students the opportunity to develop knowledge, understanding and skills in software engineering, hardware and software integration, and in the development, implementation and evaluation of computer programs. Students focus on a systematic approach to problem-solving when designing and developing creative software solutions.

What students learn

Through the study of Software Engineering 11–12, students learn to:

- analyse the social, ethical and legal implications of software engineering on individuals, society and the environment
- evaluate practices to safely and securely collect, use and store data
- design, develop, implement and evaluate safe and secure software solutions.

Course structure and requirements

Year 11 course structure and requirements

Year 11 course (120 hours)

Year 11 focus areas	Indicative hours
Programming Fundamentals	40
The Object-Oriented Paradigm	40
Programming Mechatronics	40

Year 12 course structure and requirements

Year 12 course (120 hours)

Year 12 focus areas	Indicative hours
Secure Software Architecture	30
Programming for the Web	30
Software Automation	30
Software Engineering Project	30

Safety and risk management

Schools are required to ensure they follow [safety and risk management](#) in delivering the *Software Engineering 11–12 Syllabus*.

Software Engineering course specifications

The [Software Engineering Course Specifications](#) are an integral part of the course content for Year 11 and Year 12 and indicate the depth of study required for some concepts in the *Software Engineering 11–12 Syllabus*. The *Software Engineering 11–12 Syllabus* must be applied in conjunction with the Software Engineering Course Specifications.

Course enrolment details

Software Engineering Year 11

- Course number: 11345
- Course hours: 120
- Course units: 2
- Enrolment type: Elective
- Endorsement type: Board developed

Exclusions

- Computing Technology Life Skills (Year 11, 2 units): 16623
- Technology Life Skills (Year 11, 2 units): 16686

Prerequisites

- Nil

Corequisites

- Nil

Software Engineering Year 12

- Course number: 15365
- Course hours: 120
- Course units: 2
- Enrolment type: Elective
- Endorsement type: Board developed

Exclusions

- Computing Technology Life Skills (Year 12, 2 units): 16623

- Technology Life Skills (Year 12, 2 units): 16686

Prerequisites

- Software Engineering (Year 11, 2 units): 11345

Corequisites

- Nil

HSC information

Information about [curriculum requirements](#) for the HSC are available on [Assessment Certification Examination \(ACE\)](#).

Rationale

The study of *Software Engineering 11–12* enables students to develop an understanding of software engineering as a facet of computer science. Students have the opportunity to develop knowledge and understanding of software engineering, hardware and software integration, and the development, implementation and evaluation of computer programs. They focus on a systematic approach to problem-solving when designing and developing creative software solutions.

Software Engineering promotes a deeper understanding of fundamental concepts, programming languages and innovative technologies, leading to greater flexibility when developing software solutions. Students perform project work and apply their knowledge and skills in: programming fundamentals, the object-oriented paradigm, programming mechatronics, secure software architecture, programming for the web and software automation, and use the acquired knowledge and skills to develop a software engineering project. Project work enables students to collaborate on problems and develop team and communication skills that are highly valued in the industry.

Software Engineering encourages students to explore the impact of innovations in computing technology on society and the environment. They engage with technologies that improve access to, and participation in, a range of industries.

The *Software Engineering 11–12 Syllabus* provides students with the opportunity to develop their computing skills across 4 domains: technical skills, social awareness, project management and thinking skills. Students are encouraged to transfer knowledge to new situations and projects, building on technical skills and past learning. They enhance their understanding of project management through collaboration, communicating ideas, engaging in processes and designing solutions.

Aim

The aim of Software Engineering is to develop in each student:

- a capacity to think creatively to develop and program software solutions
- an ability to apply knowledge, understanding and thinking skills to develop and communicate solutions to real-world problems.

Table of outcomes

Year 11	Year 12
SE-11-01 describes methods used to plan, develop and engineer software solutions	SE-12-01 justifies methods used to plan, develop and engineer software solutions
SE-11-02 explains how structural elements are used to develop programming code	SE-12-02 applies structural elements to develop programming code
SE-11-03 describes how current hardware, software and emerging technologies influence the development of software engineering solutions	SE-12-03 analyses how current hardware, software and emerging technologies influence the development of software engineering solutions
SE-11-04 applies safe and secure practices to collect, use and store data	SE-12-04 evaluates practices to safely and securely collect, use and store data
SE-11-05 describes the social, ethical and legal implications of software engineering on the individual, society and the environment	SE-12-05 explains the social, ethical and legal implications of software engineering on the individual, society and the environment
SE-11-06 applies tools and resources to design, develop, manage and evaluate software	SE-12-06 justifies the selection and use of tools and resources to design, develop, manage and evaluate software
SE-11-07 implements safe and secure programming solutions	SE-12-07 designs, develops and implements safe and secure programming solutions
SE-11-08 applies language structures to refine code	SE-12-08 tests and evaluates language structures to refine code
SE-11-09 manages and documents the development of a software project	SE-12-09 applies methods to manage and document the development of a software project

Outcomes and content for Year 11

Programming fundamentals

Outcomes

A student:

- describes methods used to plan, develop and engineer software solutions **SE-11-01**
- explains how structural elements are used to develop programming code **SE-11-02**
- describes how current hardware, software and emerging technologies influence the development of software engineering solutions **SE-11-03**
- applies safe and secure practices to collect, use and store data **SE-11-04**
- applies tools and resources to design, develop, manage and evaluate software **SE-11-06**
- implements safe and secure programming solutions **SE-11-07**

Content

Software development

- Explore fundamental software development steps used by programmers when designing software

Including:

- requirements definition
- determining specifications
- design
- development
- integration
- testing and debugging
- installation
- maintenance

- Research and [evaluate](#) the prevalence and use of online code collaboration tools

Designing algorithms

- Apply [computational thinking](#) and algorithmic design by defining the key features of standard [algorithms](#), including sequence, selection, iteration and identifying [data](#) that should be stored
- Apply divide and conquer and backtracking as algorithmic design strategies
- Develop structured algorithms using [pseudocode](#) and flowcharts, including the use of subprograms
- Use modelling tools including structure charts, abstraction and refinement diagrams to support top-down and bottom-up design
- Analyse the logic and structure of written algorithms

Including:

- determining inputs and outputs
- determining the purpose of the algorithm
- desk checking and peer checking
- determining connections of written algorithms to other subroutines or functions

- Identify procedures and functions in an algorithm

- Experiment with [object-oriented programming](#), imperative, logic and functional programming paradigms

Data for software engineering

- Investigate the use of number systems for computing purposes, including binary, decimal and hexadecimal
- Represent integers using two's complement
- Investigate standard [data types](#)

Including:

- char (character) and string
- Boolean
- real
- single precision floating point
- integer
- date and time

- Create data dictionaries as a tool to describe data and data types, structure data, and record relationships
- Use data structures of arrays, records, trees and sequential files

Developing solutions with code

- Apply skills in computational thinking and programming to develop a software solution

Including:

- converting an algorithm into code
- using control structures
- using data structures
- using standard modules
- creating relevant subprograms that incorporate parameter passing

- Implement data structures that support data storage

Including:

- single and multidimensional arrays
- lists
- trees
- stacks
- hash tables

- Compare the execution of the Waterfall and Agile project management [models](#) as applied to software development
- Test and evaluate solutions, considering key aspects including functionality, performance, readability of code, quality of documentation
- Use debugging tools

Including:

- breakpoints
- single line stepping
- watches
- interfaces between functions
- debugging output statements
- debugging software available in an integrated development environment (IDE)

- Determine sets of suitable test data

Including:

- boundary values
- path coverage
- faulty and abnormal data

- Determine typical errors experienced when developing code, including syntax, logic and runtime, and explain their likely causes

The object-oriented paradigm

Outcomes

A student:

- describes methods used to plan, develop and engineer software solutions **SE-11-01**
- explains how structural elements are used to develop programming code **SE-11-02**
- describes how current hardware, software and emerging technologies influence the development of software engineering solutions **SE-11-03**
- applies safe and secure practices to collect, use and store data **SE-11-04**
- applies tools and resources to design, develop, manage and evaluate software **SE-11-06**
- implements safe and secure programming solutions **SE-11-07**
- applies language structures to refine code **SE-11-08**
- manages and documents the development of a software project **SE-11-09**

Content

Understanding OOP

- Apply the key features of an [object-oriented programming \(OOP\)](#) language

Including:

- objects
- classes
- encapsulation
- abstraction
- inheritance
- generalisation
- polymorphism

- Compare [procedural programming](#) with OOP
- Use [data](#) flow diagrams, structure charts and class diagrams to represent a system
- Describe the process of design used to develop code in an OOP language

Including:

- task definition
- top-down and bottom-up
- facade pattern
- agility

- Assess the effectiveness of programming code developed to implement an [algorithm](#)
- Investigate how OOP languages handle message-passing between objects
- Explain code optimisation in software engineering
- Outline the features of OOP that support collaborative code development

Including:

- consistency
- code commenting
- version control
- feedback

Programming in OOP

- Design and implement computer programs involving branching, iteration and functions in an OOP language for an identified need or opportunity
- Implement and modify OOP programming code

Including:

- clear and uncluttered mainline
- one logical task per subroutine
- use of stubs
- use of control structures and data structures
- ease of maintenance
- version control
- regular backup

- Apply methodologies to test and [evaluate](#) code

Including:

- unit, subsystem and system testing
- black, white and grey box testing
- quality assurance

Programming mechatronics

Outcomes

A student:

- describes methods used to plan, develop and engineer software solutions **SE-11-01**
- explains how structural elements are used to develop programming code **SE-11-02**
- describes how current hardware, software and emerging technologies influence the development of software engineering solutions **SE-11-03**
- applies safe and secure practices to collect, use and store data **SE-11-04**
- describes the social, ethical and legal implications of software engineering on the individual, society and the environment **SE-11-05**
- applies tools and resources to design, develop, manage and evaluate software **SE-11-06**
- implements safe and secure programming solutions **SE-11-07**
- applies language structures to refine code **SE-11-08**
- manages and documents the development of a software project **SE-11-09**

Content

Understanding mechatronic hardware and software

- Outline applications of mechatronic systems in a variety of specialised fields
- Identify the hardware requirements to run a program and the effect on code development

Including:

- assessing the relationship of microcontrollers and the central processing unit (CPU)
- the influence of instruction set and opcodes
- the use of address and data registers

- Identify and describe a range of sensors, actuators and end effectors/manipulators within existing mechatronic systems

Including:

- motion sensors
- light level sensors
- hydraulic actuators
- robotic grippers

- Use different types of [data](#) and understand how it is obtained and processed in a mechatronic system, including diagnostic data and data used for optimisation
- Experiment with software to control interactions and dependencies within mechatronic systems

Including:

- motion constraints
- degrees of freedom
- combination of subsystems
- combination of sensors, actuators and end effectors to create viable subsystems

- Determine power, battery and material requirements for components of a mechatronic system
- Develop wiring diagrams for a mechatronic system, considering data and power supply requirements
- Determine specialist requirements that influence the design and functions of mechatronic systems designed for people with [disability](#)

Designing control algorithms

- Develop, modify and apply [algorithms](#) to control a mechatronic system
- Explore the algorithmic patterns, code and applications for open and closed control systems
- Outline the features of an algorithm and program code used for autonomous control

Programming and building

- Design, develop and produce a mechatronic system for a real-world problem

Including:

- software control
 - mechanical engineering
 - electronics and mathematics
- Implement algorithms and design programming code to drive mechatronic devices
 - Develop simulations and [prototypes](#) of a potential mechatronic system to test programming code
 - Design, develop and implement programming code for a closed loop control system
 - Apply programming code to integrate sensors, actuators and end effectors/manipulators
 - Implement specific control algorithms that enhance the performance of a mechatronic system
 - Design, develop and implement a [user interface \(UI\)](#) to control a mechatronic system
 - Create and use unit tests to determine the effectiveness and repeatability of each component's control algorithm

Outcomes and content for Year 12

Secure software architecture

Outcomes

A student:

- justifies methods used to plan, develop and engineer software solutions **SE-12-01**
- applies structural elements to develop programming code **SE-12-02**
- analyses how current hardware, software and emerging technologies influence the development of software engineering solutions **SE-12-03**
- evaluates practices to safely and securely collect, use and store data **SE-12-04**
- explains the social, ethical and legal implications of software engineering on the individual, society and the environment **SE-12-05**
- justifies the selection and use of tools and resources to design, develop, manage and evaluate software **SE-12-06**
- designs, develops and implements safe and secure programming solutions **SE-12-07**
- tests and evaluates language structures to refine code **SE-12-08**

Content

Designing software

- Describe the benefits of developing secure software

Including:

- data protection
- minimising cyber attacks and vulnerabilities

- Interpret and apply fundamental software development steps to develop secure code

Including:

- requirements definition
- determining specifications
- design
- development
- integration
- testing and debugging
- installation
- maintenance

- Describe how the capabilities and experience of end users influence the secure design features of software

Developing secure code

- Explore fundamental software design security concepts when developing programming code

Including:

- confidentiality
- integrity
- availability
- authentication
- authorisation
- accountability

- Apply security features incorporated into software including [data](#) protection, security, privacy and regulatory compliance
- Use and explain the contribution of cryptography and sandboxing to the 'security by design' approach in the development of software solutions
- Use and explain the 'privacy by design' approach in the development of software solutions

Including:

- proactive not reactive approach
- embed privacy into design
- respect for user privacy

- Test and [evaluate](#) the security and resilience of software by determining vulnerabilities, hardening systems, handling breaches, maintaining business continuity and conducting disaster recovery
- Apply and evaluate strategies used by software developers to manage the security of programming code

Including:

- code review
- static application security testing (SAST)
- dynamic application security testing (DAST)
- vulnerability assessment
- penetration testing

- Design, develop and implement code using defensive data input handling practices, including input validation, sanitisation and error handling
- Design, develop and implement a safe [application programming interface \(API\)](#) to minimise software vulnerabilities
- Design, develop and implement code considering efficient execution for the user

Including:

- memory management
- session management
- exception management

- Design, develop and implement secure code to minimise vulnerabilities in user action controls

Including:

- broken authentication and session management
- cross-site scripting (XSS) and cross-site request forgery (CSRF)
- invalid forwarding and redirecting
- race conditions

- Design, develop and implement secure code to protect user file and hardware vulnerabilities from file attacks and side channel attacks

Impact of safe and secure software development

- Apply and describe the benefits of collaboration to develop safe and secure software

Including:

- considering various points of view
- delegating tasks based on expertise
- quality of the solution

- Investigate and explain the benefits to an enterprise of the implementation of safe and secure development practices

Including:

- improved products or services
- influence on future software development
- improved work practices
- productivity
- business interactivity

- Evaluate the social, ethical and legal issues and ramifications that affect people and enterprises resulting from the development and implementation of safe and secure software

Including:

- employment
- data security
- privacy
- copyright
- intellectual property
- digital disruption

Programming for the web

Outcomes

A student:

- justifies methods used to plan, develop and engineer software solutions **SE-12-01**
- applies structural elements to develop programming code **SE-12-02**
- analyses how current hardware, software and emerging technologies influence the development of software engineering solutions **SE-12-03**
- evaluates practices to safely and securely collect, use and store data **SE-12-04**
- explains the social, ethical and legal implications of software engineering on the individual, society and the environment **SE-12-05**
- justifies the selection and use of tools and resources to design, develop, manage and evaluate software **SE-12-06**
- designs, develops and implements safe and secure programming solutions **SE-12-07**
- tests and evaluates language structures to refine code **SE-12-08**
- applies methods to manage and document the development of a software project **SE-12-09**

Content

Data transmission using the web

- Explore the applications of web programming

Including:

 - interactive website/webpages
 - e-commerce
 - progressive web apps (PWAs)
- Investigate and practise how [data](#) is transferred on the internet

Including:

 - data packets
 - internet protocol (IP) addresses, including IPv4
 - domain name systems (DNS)
- Investigate and describe the function of web [protocols](#) and their ports

Including:

 - HTTP, HTTPS
 - TCP/IP
 - DNS
 - FTP, SFTP
 - SSL, TLS
 - SMTP, POP 3, IMAP
- Explain the processes for securing the web

Including:

- Secure Sockets Layer (SSL) certificates
- encryption algorithms
- encryption keys
- plain text and cipher text
- authentication and authorisation
- hash values
- digital signatures

- Investigate the effect of [big data](#) on web architecture

Including:

- data mining
- metadata
- streaming service management

Designing web applications

- Investigate and explain the role of the World Wide Web Consortium ([W3C](#)) in the development of applications for the web

Including:

- Web Accessibility Initiative (WAI)
- internationalisation
- web security
- privacy
- machine-readable data

- Model elements that form a web development system

Including:

- client-side (front-end) web programming
- server-side (back-end) web programming
- interfacing with databases that are based on Structured Query Language (SQL) or non-SQL

- Explore and explain the influence of a web browser on web development, including the use of developer (dev) tools
- Investigate cascading style sheets (CSS) and its impact on the design of a web application

Including:

- consistency of appearance
- flexibility with browsers or display devices
- CSS maintenance tools

- Investigate the reasons for version control and apply it when developing web application
- Explore the types and significance of code libraries for front-end web development

Including:

- frameworks that control complex web applications
- template engines
- predesigned CSS classes

- Explain the use and development of open-source software in relation to web development
- Investigate methods to support and manage the load times of web pages/applications

- Research, experiment with and [evaluate](#) the prevalence and use of web content management systems (CMS)
- Assess the contribution of back-end web development to the success of a web application
- Observe and describe the back-end process used to manage a web request

Including:

- role of webserver software
- web framework
- objects
- libraries
- databases

- Develop a web application using an appropriate [scripting language](#) with shell scripts to make files and directories, and searching for text in a text file
- Apply a web-based database and construct script that executes SQL

Including:

- selecting fields
- incorporating 'group by'
- common SQL queries
- constraints using WHERE keyword
- table joins

- Compare [Object-Relational Mapping \(ORM\)](#) to SQL
- Describe how collaborative work practices between front-end and back-end developers improve the development of a web solution
- Design, develop and implement a progressive web app (PWA)

Including:

- the application of design and user interface (UI) and user experience (UX) principles of font, colour, audio, video and navigation
- a UI that considers accessibility and inclusivity

Software automation

Outcomes

A student:

- justifies methods used to plan, develop and engineer software solutions **SE-12-01**
- applies structural elements to develop programming code **SE-12-02**
- analyses how current hardware, software and emerging technologies influence the development of software engineering solutions **SE-12-03**
- evaluates practices to safely and securely collect, use and store data **SE-12-04**
- explains the social, ethical and legal implications of software engineering on the individual, society and the environment **SE-12-05**
- justifies the selection and use of tools and resources to design, develop, manage and evaluate software **SE-12-06**
- designs, develops and implements safe and secure programming solutions **SE-12-07**
- tests and evaluates language structures to refine code **SE-12-08**
- applies methods to manage and document the development of a software project **SE-12-09**

Content

Algorithms in machine learning

- Investigate how machine learning (ML) supports automation through the use of DevOps, robotic process automation (RPA) and business process automation (BPA)
- Distinguish between [artificial intelligence \(AI\)](#) and ML
- Explore [models](#) of training ML

Including:

- supervised learning
- unsupervised learning
- semi-supervised learning
- reinforcement learning

- Investigate common applications of key ML [algorithms](#)

Including:

- data analysis and forecasting
- virtual personal assistants
- image recognition

- Research models used by software engineers to design and analyse ML

Including:

- decision trees
- neural networks

- Describe types of algorithms associated with ML

Including:

- linear regression
- logistic regression
- K-nearest neighbour

Programming for automation

- Design, develop and apply ML regression models using an [OOP](#) to predict numeric values

Including:

- linear regression
- polynomial regression
- logistic regression

- Apply neural network models using an OOP to make predictions

Significance and impact of ML and AI

- Assess the impact of automation on the individual, society and the environment

Including:

- safety of workers
- people with disability
- the nature and skills required for employment
- production efficiency, waste and the environment
- the economy and distribution of wealth

- Explore by implementation how patterns in human behaviour influence ML and AI software development

Including:

- psychological responses
- patterns related to acute stress response
- cultural protocols
- belief systems

- Investigate the effect of human and dataset source bias in the development of ML and AI solutions

Software engineering project

Outcomes

A student:

- justifies methods used to plan, develop and engineer software solutions **SE-12-01**
- applies structural elements to develop programming code **SE-12-02**
- analyses how current hardware, software and emerging technologies influence the development of software engineering solutions **SE-12-03**
- evaluates practices to safely and securely collect, use and store data **SE-12-04**
- explains the social, ethical and legal implications of software engineering on the individual, society and the environment **SE-12-05**
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- designs, develops and implements safe and secure programming solutions **SE-12-07**
- tests and evaluates language structures to refine code **SE-12-08**
- applies methods to manage and document the development of a software project **SE-12-09**

Content

Identifying and defining

- Define and analyse the requirements of a problem

Including:

- demonstrating need(s) or opportunities
- assessing scheduling and financial feasibility
- generating requirements including functionality and performance
- defining data structures and data types
- defining boundaries

- Explore tools used to develop ideas and generate solutions

Including:

- brainstorming, mind-mapping and storyboards
- data dictionaries, including selecting appropriate data types
- algorithm design
- code generation
- testing and debugging
- installation
- maintenance

- Investigate types of software implementation methods

Including:

- direct
- phased
- parallel
- pilot

Research and planning

- Research and use the Waterfall software development approach

Including:

- logical progression of steps used throughout the life cycle
- stages of 'falling water'
- advantages and disadvantages
- scale and types of developments

- Research and use the Agile software development approach

Including:

- rate of developing a final solution
- method tailoring
- iteration workflow
- scale and types of developments

- Research the WAgile software development approach

Including:

- understanding it is a hybrid model
- analysis of the 'when' and 'how' intervention is applied during the development life cycle
- scale and types of developments

- Apply project management to plan and conduct the development and implementation of a project and software engineering solution

Including:

- scheduling and tracking using a software tool, including Gantt charts
- using collaboration tools

- Explore social and ethical issues associated with project work, including working individually, collaboratively and responding to stakeholders
- Explore communication issues associated with project work

Including:

- involving and empowering the client
- enabling feedback
- negotiating

- Investigate how software engineering solutions are quality assured

Including:

- defining criteria on which quality will be judged
- ensuring requirements are met using a continual checking process
- addressing compliance and legislative requirements

- Demonstrate the use of modelling tools
- Explain the contribution of back-end engineering to the success and ease of software development

Including:

- technology used
- error handling
- interfacing with front end
- security engineering

Producing and implementing

- Design, construct and implement a solution to a software problem using appropriate development approach(es)
- Present a software engineering solution using presentation software
- Develop, construct and document [algorithms](#)
- Allocate resources to support the development of a software engineering solution
- Demonstrate the use of programmed [data](#) backup
- Implement version control when developing a software engineering solution
- Explore strategies to respond to difficulties when developing a software engineering solution

Including:

- looking for a solution online
- collaboration with peers
- outsourcing

- Propose an additional innovative solution using a [prototype](#) and [user interface \(UI\)](#) design

Testing and evaluating

- Apply methodologies to test and [evaluate](#) code
- Use a language-dependent code optimisation technique
- Analyse and respond to feedback
- Evaluate the effectiveness of a software engineering solution

Including:

- developing a report to synthesise feedback
- developing a test plan
- testing data used/generated based on path and boundary testing
- comparing actual output with expected output

Assessment

Course standards

Course standards are comprised of syllabus standards and performance standards. Syllabus outcomes and content form the syllabus standards that describe what students are expected to know, understand and do. Performance standards, including the Common Grade Scale for Preliminary courses, HSC Performance Band Descriptions and Achievement Level Descriptions (for HSC courses with grades only) describe how well students have achieved in relation to syllabus standards. Performance standards are used to report student achievement on NESA credentials.

Common Grade Scale for Preliminary courses

Stage 6 – Year 11

The [Common Grade Scale for Preliminary courses](#) provides generic, holistic descriptions for performance at each of the 5 grade levels. It is used to report student achievement in Year 11 courses in all NSW schools.

Performance Band Descriptions for Software Engineering

Stage 6 – Year 12

HSC performance band descriptions summarise the knowledge, understanding and skills typically demonstrated by students at each band level in a particular course. NESA uses these holistic descriptions to report a student's overall level of achievement in an HSC course.

Band 6

A student who performs at this level typically:

- demonstrates comprehensive understanding of the influence of technologies on the development of software engineering solutions
- demonstrates extensive knowledge and understanding of the social, ethical and legal implications of the application of software engineering solutions on the individual, society and the environment
- applies sophisticated technical skills in the design and implementation of programming solutions
- collects, uses, manages and stores complex data effectively, safely and securely
- applies comprehensive skills in developing, managing and documenting software engineering projects
- communicates logically and effectively using a range of terms, conventions and methods.

Band 5

A student who performs at this level typically:

- demonstrates broad understanding of the influence of technologies on the development of software engineering solutions
- demonstrates thorough knowledge and understanding of the social, ethical and legal implications of the application of software engineering solutions on the individual, society and the environment
- applies high-level technical skills in the design and implementation of programming solutions
- collects, uses, manages and stores data effectively, safely and securely

- demonstrates high-level skills in developing, managing and documenting software engineering projects
- communicates logically, using appropriate terms, conventions and methods.

Band 4

A student who performs at this level typically:

- demonstrates sound understanding of the influence of technologies on the development of software engineering solutions
- demonstrates sound knowledge and understanding of the social, ethical and legal implications of the application of software engineering solutions
- applies sound technical skills in the design and implementation of programming solutions
- collects, uses, manages and stores data safely and securely
- demonstrates sound skills in developing, managing and documenting software engineering projects
- communicates using relevant terms, conventions and methods.

Band 3

A student who performs at this level typically:

- demonstrates basic understanding of the influence of technologies on the development of software engineering solutions
- demonstrates basic knowledge and understanding of the social and/or ethical and/or legal implications of the application of software engineering solutions
- displays basic skills in programming
- uses data safely and securely
- displays basic skills in developing software engineering projects
- communicates using basic terms and conventions and/or methods.

Band 2

A student who performs at this level typically:

- demonstrates limited understanding of the use of technologies
- displays limited skills in programming
- displays limited skills in developing software engineering projects
- communicates with limited use of terms.

Band 1

Performance reported as Band 1 indicates that the minimum standard expected was not demonstrated.

School-based assessment

Schools are required to develop an assessment program for each Year 11 and Year 12 course. NESAs provides information about the responsibilities of schools in developing assessment programs in course-specific assessment requirements and in the Assessment Certification Examination (ACE) rules and requirements.

Year 11 Software Engineering school-based assessment

Stage 6 – Year 11

Schools are required to submit to NESAs a grade for each student based on their achievement at the end of the course.

Teachers use professional, on-balance judgement to allocate grades based on the Common Grade Scale for Preliminary courses.

Teachers consider all available assessment information, including formal and informal assessment, to determine the grade that best matches each student's achievement at the end of the course.

See [Assessment Certification Examination \(ACE\) rules and requirements](#) for further information.

Assessment programs must reflect course components and weightings

The course components and component weightings for Year 11 are mandatory.

Course component	Weighting %
Knowledge and understanding of course content	50
Knowledge and skills in the practical application of the content	50

Schools may determine specific elements of their assessment program

Schools have authority to determine the number, type of task and the weighting allocated to an assessment task. Schools may also follow the sample assessment programs provided by NESA.

Sample assessment program

NESA's sample Year 11 formal school-based assessment program for Software Engineering is:

- 3 assessment tasks, including:
 - a written examination
 - a task based on a project
 - a weighting for any individual task of 20% to 40%.

Some students with disability may require [adjustments](#) in order to access assessment opportunities and demonstrate achievement of outcomes.

Year 12 Software Engineering school-based assessment

Stage 6 – Year 12

NESA requires schools to submit a school-based assessment mark for each Year 12 student who completes this course. Formal school-based assessment tasks should reflect the syllabus outcomes and content. The mark submitted by the school provides a summation of each student's achievement measured at several points throughout the course. A school's program of school-based assessment includes both mandatory and non-mandatory elements.

See [Assessment Certification Examination \(ACE\) rules and requirements](#) for further information.

Assessment programs must reflect course components and weightings

The course components and component weightings for Year 12 are mandatory.

Course component	Weighting %
Knowledge and understanding of course content	50
Knowledge and skills in the practical application of the content	50

Schools may determine specific elements of their assessment program

Schools have authority to determine the number, type of task and the weighting allocated to an assessment task. Schools may also follow the sample assessment programs provided by NESA.

Sample assessment program

NESA's sample Year 12 formal school-based assessment program for Software Engineering is:

- 4 assessment tasks, including:
 - an individual project with a weighting of 30%
 - a formal examination
 - a weighting for any individual task of 10% to 40%.

Some students with disability may require [adjustments](#) in order to access assessment opportunities and demonstrate achievement of outcomes.

Sample assessment schedules

Stage 6

- [Sample assessment schedule Year 11](#)
- [Sample assessment schedule Year 12](#)

Sample assessment task notifications

Stage 6

- [Sample formal assessment task notification: Year 11 \(Mechatronic programming project\)](#)
- [Sample formal assessment task notification: Year 12 \(Software engineering project\)](#)

HSC examinations

Examination questions may require students to integrate knowledge, understanding and skills developed through studying the course.

The Year 11 course is assumed knowledge for the Year 12 course.

An understanding of the Python programming language is also assumed.

The examination will be worth 80 marks.

It will be undertaken by students using a computer.

The time allowed is 2 hours and 30 minutes including 10 minutes reading time.

Questions may contain stimulus material.

Headphones will be required for questions with video and audio stimulus.

The examination will consist of two types of test item.

Some students with disability may be eligible for [HSC exam provisions](#) when completing the HSC examination.

Software Engineering HSC examination specifications

Stage 6 – Year 12

Objective-response items (approx. 20 marks)

- Objective-response items may share stimulus with other objective-response items or short-answer items.
- Each item will be worth 1 to 4 marks.
- There will be 14 to 18 items in total.

Short-answer items (approx. 60 marks)

- Short-answer items may share stimulus with other short-answer items or objective-response items.
- At least three items will be worth 4 to 8 marks.
- There will be 16 to 18 items in total.

Features of online HSC exams

Stage 6 – Year 12

To learn more about the format and features of online HSC exams, see [our collection of videos](#).

Tools available in the Software Engineering online exam

There are specific tools available for this online exam as demonstrated in these videos.

- [Code editor](#)
- [Code editor video transcript](#)
- [Drawing tool](#)
- [Drawing tool video transcript](#)
- [Screen design tool](#)
- [Screen design tool video transcript](#)

HSC examination materials

Stage 6 – Year 12

An online sample exam for Software Engineering 11–12 is now available. Schools can enrol students in the sample exam via [Schools Online](#). A guide on how to enrol students and conduct the exams is available in Schools Online > Downloads > Memos and Documents > HSC Online Exam Resources. Schools can determine the most appropriate time to enrol students and for the students to sit the sample exams.

The marking guidelines for the online sample exam are also available.

- [HSC familiarisation questions](#)
- [Marking guidelines for HSC familiarisation questions](#)
- [Marking guidelines for the online HSC sample exam](#)
- [Past HSC exam papers archive \(see Software Design and Development\)](#)
- [Software Engineering HSC exam papers](#)

Further information

Stage 6 – Year 12

- [Assessment in Stage 6](#)
- [Principles of effective assessment](#)
- [Assessment in practice](#)
- [School-based assessment requirements](#)
- [Adjustments to assessment for students with disability](#)
- [Life Skills](#)
- [Assessment Certification Examination \(ACE\) website](#)

